

PEAS DESCRIPTION OF TASK ENVIRONMENTS

Tasks	Performance Measure	Environment	Actuators	Sensors
Self Driving Car Navigation	Safety, arrival on time, following traffic rules, no collisions	roads, cars and pedestrians around, speed breakers, road blocks, dividers	Steering, brake, throttle, gears	Cameras, proximity sensors, real time navigation support
Autonomous Drone Delivery	Arrival on time, no collisions with houses, accuracy of delivery, safety of operation	the sky, the birds, airspace, weather	Rotors, motors, delivery mechanism	Cameras, proximity sensors, real time navigation support
Smart Home Vacuum Cleaning Robot	Clean surroundings, no breakdown of the machinery due to long usage, time cleaning, coverage of the area	Rooms, floors, carpets, furniture	A suction pipe, a motor, an exhaust fan, container to catch the dirt, brushes	Cameras, proximity sensors
Online Shopping Recommendation System	User satisfaction, increased sales	The products to sell, the customer details, the past order history of the user and other users	Recommendation display on user interface	User profile, past behavior, product data
Stock Market Trading Agent	Uplift in the portfolio, sustained profits over a long period of time	the everchanging stock market	stock market operations such as buy, sell, hold	the real time stock market data, news updates
Intelligent Personal Agent	Effective and efficient communication between the user and the agent, execution of the commands given by the user, user satisfaction, accuracy of the responses	User queries, the user interface to operate within	Voice/text responses, actions on apps, web access	Microphone, cameras, user clicks and accesses
Autonomous Mars Rover Exploration	Successful exploration of the planet, no breakdown of the machinery, report of the data collected	Mars' environment	wheels, arms of the robot, landing shuttle, launch pods	Cameras, proximity sensors, real time navigation support

Email Spam Filter	Elimination of Spams, successful delivery of non-spam emails	The mailbox of the user, the spambox	Mark email as spam or not spam	Email body, subject, sender info
Medical Diagnosis System	High accuracy, low false positives and false negatives	Patient data, medical information	Diagnosis output, treatment suggestion	Patient symptoms, test results, medical history
AI Powered Chess Playing Agent	Winning the game, correct prediction of the opponent's moves, making accurate and smart moves	the chessboard	Move chess pieces on the board	Opponent's move input, board state
Automated Language Translation System	Accurate, reliable and speedy translation of the input user language	Input text/audio	Translated text/audio output	Input language data
Face Recognition Security System	Accurate, reliable and speedy face recognition, updating user data in case of no match	Faces of people at entry point	Grant or deny access	Camera, face recognition software
Traffic Signal Control System	Efficient and effective control of road traffic, prevention of collisions, safety of the commuters	the roads, pedestrians, vehicles operating on the roads, other traffic signals	Traffic lights	Camera, real time access to the traffic navigation systems
Autonomous Warehouse Robot	Accurate, speedy and efficient warehouse management	the warehouse, the products, shelves	wheels, arms of the robot, elevators	Cameras, proximity sensors, real time product and shelf movement data
AI Based Tutoring System	Improved student learning outcomes, delivery of accurate material	Students, coursework, tests	Feedback, hints, explanations, assessments	Student responses, performance metrics
Voice-to-Text Transcription System	Accurate, speedy and reliable transcription	the user audio input	Text output	Microphone
Autonomous Farming Robot	Successful farming outputs and higher yields	Soil, fields, crops, weather	Robotic arms, harvestors, tillers, sowing machines, wheels	Camera, proximity sensors, soil depth sensors

AI News Article Summarizer	Real time delivery of accurate and reliable news	News articles from online sources	Summarized text output	News article text input
AI Game Non-Player Behaviour Control	Realistic movement of the NPC, no unnecessary interaction with the main character	Real time environment in the game, player's action	Movement, interaction with the player and other NPC's	Cameras, sensors, game data
AI Legal Document Analyzer	Accurate, reliable analysis of the legal documents	The legal document, the domain of the legality, user case data	Analysis report, suggestions and advice	Legal documents, historical case data